

Security Dashboard using Microsoft Sentinel (Local Simulation)

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Outline

- 1 Introduction
- 2 Literature Review
- 3 Research Gaps Identified
- 4 Problem Statement
- 5 Objectives
- 6 Proposed System
- 7 Admin Authentication Flow
- 8 Proposed System
- 9 Security Dashboard Working
- 10 Output
- 11 Expected Outcomes
- 12  Conclusion

Introduction

- Increasing cyber threats and unauthorized access attempts demand stronger authentication and monitoring mechanisms.
- Traditional password-based authentication systems are insufficient for protecting admin-level applications.
- Secure authentication requires identity verification, behavior analysis, and strict access control mechanisms.
- This project focuses on developing a secure admin authentication system with multiple security layers.
- The authentication process begins with username and password verification for initial access control.



Introduction (contd.)

- JWT-based authentication is used to securely manage user sessions without maintaining server-side session storage.
- Multi-Factor Authentication (MFA) using One-Time Passwords (OTP) adds an additional identity verification layer.
- IP whitelisting ensures admin access is restricted to trusted network locations.
- All login attempts are logged and analyzed to detect suspicious behavior such as bot activity and repeated failures.
- A security dashboard visualizes real-time metrics, calculates risk scores, and enables flagging or blocking of high-risk users or IP addresses.



Literature Review

Author & Year	Method	Result	Future Scope
Krzysztof Goczyła , 2010 [1]	Introduced KQL as a SQL-like query language for knowledge bases.	Improved modular reasoning and data representation.	Extend KQL for numeric data and access control.
C.W. Johnson, 2014 [2]	Combined safety and security reporting using forensic analysis.	Identified challenges in cyber incident reporting for safety-critical systems.	Improve automation and cooperation in forensic data sharing.
Igor Anastasov, Danco Davcev, 2014 [3]	SIEM model using hierarchical management and ontology repository.	Better log management and attack correlation.	Integrate AI for adaptive event detection.
Kai-Oliver Detken , 2015 [4]	Developed lightweight SIEM (SIMU project) using CBOR and IF-MAP.	Enhanced IT security for SME networks.	Expand to handle distributed network monitoring.
Kemal Delic, Jeff Riley, 2015 [5]	Smart Log Analytics using Fuzzy Logic, Decision Trees, and Neural Nets.	Automated, cost-efficient log analysis.	Extend with AI for large-scale cloud-based systems.



Literature Review (contd.)

Ana Vazão , 2019 [6]	Compared open-source SIEM tools for performance and GDPR compliance.	No single tool satisfied all needs.	Add ML and anomaly detection features.
Ankita Verma , 2019 [7]	Azure platform and cognitive services study.	Enhanced performance in cloud-based analytics.	Expand AI capabilities in Azure environments.
Kennedy A. Torkura , 2019 [8]	Automated multi-cloud threat detection using Slingshot system.	Improved response time and reduced detection delay.	Extend to hybrid multi-cloud deployments.
Oskars Podzins, Andrejs Romanovs, 2019 [9]	Practical SIEM application for secure IT environments.	SIEM proven essential but costly.	Reduce operational cost and improve automation.
Pavel Yermalovich, 2020 [10]	Visualization techniques for security dashboards.	Enhanced decision-making and user understanding.	Develop interactive and adaptive visual tools.



Literature Review (contd.)

Andreas Schreiber , 2021 [11]	Visual analytics dashboards for security testing tools.	Improved usability and reduced false positives.	Add filtering, search, and interactive exploration.
Lukasz Korzeniowski, Krzysztof Goczyła, 2022 [12]	Systematic review of automated log analysis methods.	Identified major research gaps and challenges.	Develop unified log parsing and analysis frameworks.
Zarrin Tasnim Sworna , 2023 [13]	IRP2API model mapping incident response to security APIs.	Achieved higher accuracy in automated mappings.	Expand datasets and include contextual metadata.
Usha Desai , 2024 [14]	Developed proactive cybersecurity system using Azure HoneyNet.	Improved threat detection and cloud integration.	Incorporate AI-driven alert correlation.
Yadidiah Kanaparthi , 2024 [15]	Used ML algorithms for automated log analysis.	Achieved anomaly detection for real-time logs.	Enhance scalability and integration with SIEM tools.

Figure: Literature Review



Research Gaps Identified

- High Complexity: Current SIEM tools such as Microsoft Sentinel are high in complexity and need intricate configuration and technical knowledge.
- Cost and Resource Barriers: Cloud-based SIEM platforms have high costs, ongoing internet access, and high infrastructure requirements.
- Limited Educational Accessibility: Most SIEM solutions are enterprise-oriented and not scalable for student learning or academic purposes.



Research Gaps Identified (contd.)

- Insufficiency of Local Open-Source Models: Very few light-weight, locally deployable SIEM simulations using open-source technologies are available
- Dashboard and Visualization Insufficiencies:* Dashboards today are sophisticated and not intuitive to visualize learning in real time.



Problem Statement

- Traditional password-based authentication systems are insufficient to protect admin-level applications from modern cyber threats such as credential theft and automated bot attacks.
- Many existing systems lack multi-factor authentication, risk-based analysis, and real-time monitoring of login activities.
- Unauthorized access attempts often remain undetected due to the absence of centralized login logging and behavioral analysis.
- Enterprise-grade authentication monitoring solutions are complex, costly, and difficult to deploy in academic or small-scale environments.



Problem Statement (contd.)

- There is limited availability of practical platforms for implementing and analyzing secure authentication workflows using JWT, MFA, and access control mechanisms.



Objectives

- Design a Secure Authentication Mechanism: Implement a robust admin login system using username–password verification, JWT-based session management, and multi-factor authentication.
- Enable Advanced Access Control: Apply IP whitelisting and token-based authorization to restrict access to trusted users and network locations.
- Monitor and Log Login Activities: Capture and store all login events to support auditing, analysis, and security monitoring.



Objectives (contd.)

- Detect Suspicious Behavior Using Risk Analysis: Analyze login patterns such as failed attempts, MFA failures, bot behavior, and location anomalies to compute dynamic risk scores.
- Provide Real-Time Security Visibility: Develop a security dashboard that displays risk scores, alerts, and overall system security status to enable proactive response.



Proposed System

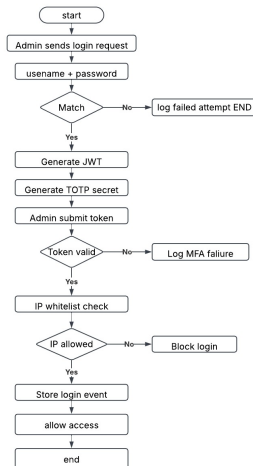


Figure: Workflow for Security Dashboard



Proposed System (contd.)



Admin Authentication and MFA Process

- Login Request Initiation

- Admin opens the login page and submits login credentials.
- A login request is sent to the server for authentication.

- Credential Verification

- Server verifies the username and password against the database.

- Invalid Credential Handling

- If credentials are incorrect, access is denied.
- Prevents unauthorized system entry.

- Temporary JWT Generation

- A temporary JWT token is generated for further verification.
- Token does not grant full system access.



Admin Authentication and MFA Process (contd.)

- **MFA Status Check**

- System checks whether Multi-Factor Authentication (MFA) is enabled.

- **Token Generation and Delivery**

- Token is sent to the admin via authenticator app.

- **Token Submission and Verification**

- Admin submits the received Token.
- Server validates the Token for correctness and expiry.

- **Risk and Behavior Analysis**

- System checks failed attempts and bot-like behavior.
- Helps in detecting brute-force or automated attacks.



Admin Authentication and MFA Process (contd.)

- **IP Blacklisting Verification**

- Admin IP address is verified against allowed IP list.
- Unauthorized IPs are blocked.

- **Logging and Risk Score Update**

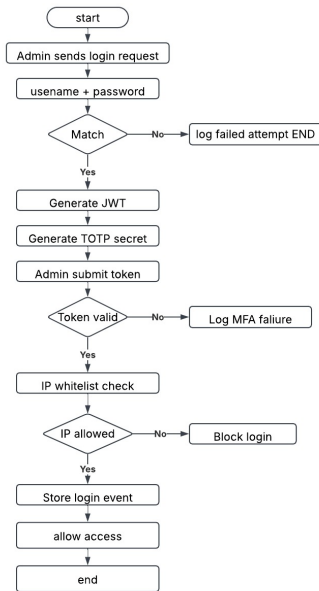
- Login activity is logged with timestamp and IP.
- User risk score is updated based on behavior.

- **Access Grant**

- Admin is granted full dashboard access after all checks pass.



Proposed System



Proposed System

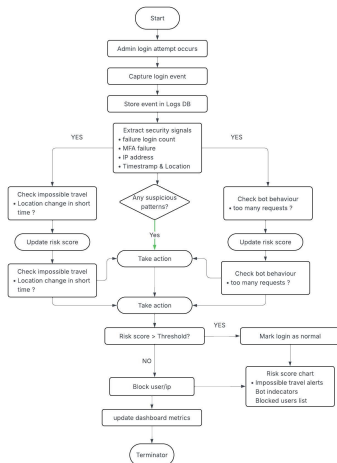


Figure: Workflow for Security Dashboard



Security Dashboard Working

- Continuous Security Monitoring

- The security dashboard continuously monitors all admin login activities to detect suspicious behavior.

- Admin Login Attempt Detection

- Every admin login attempt is automatically detected by the system.

- Login Event Capture

- The system captures and records each login event.

- Log Storage

- Login details are securely stored in the logs database for auditing and analysis.



Security Dashboard Working (contd.)

- **Security Signal Extraction**

- Security-related signals are extracted from stored login data.

- **Failed Login Analysis**

- The system tracks failed login attempts.
- Multiple failures indicate potential brute-force attacks.

- **MFA Failure Detection**

- Multi-Factor Authentication (MFA) failures are monitored.
- Repeated MFA failures increase security risk.

- **IP Address Monitoring**

- IP addresses of login requests are recorded.
- Unknown or unusual IPs are treated as suspicious.



Security Dashboard Working (contd.)

- **Time and Location Analysis**

- Login time and geographic location are analyzed.
- Helps identify abnormal login patterns.

- **Impossible Travel Detection**

- The system checks for logins from distant locations within short time intervals.
- Such cases are marked as suspicious.

- **Bot Behavior Detection**

- Detects automated behavior such as rapid repeated login attempts.
- Identifies repeated requests from the same IP address.

- **Suspicious Pattern Detection**



Security Dashboard Working (contd.)

- Login behavior is classified as normal or suspicious based on combined signals.
- **Risk Score Update**
 - Each suspicious activity increases the user's risk score.
 - Higher scores indicate greater threat levels.
- **Risk Threshold Comparison**
 - The risk score is compared against a predefined threshold.
 - Determines the system's response.
- **High-Risk Response**
 - Users or IPs exceeding the threshold are flagged or blocked.
 - Continuous monitoring is applied to high-risk entities.



Security Dashboard Working (contd.)

- **Low-Risk Handling**

- Logins below the threshold are marked as safe.
- No security action is taken.

- **Dashboard Metric Update**

- Dashboard metrics are updated in real time.

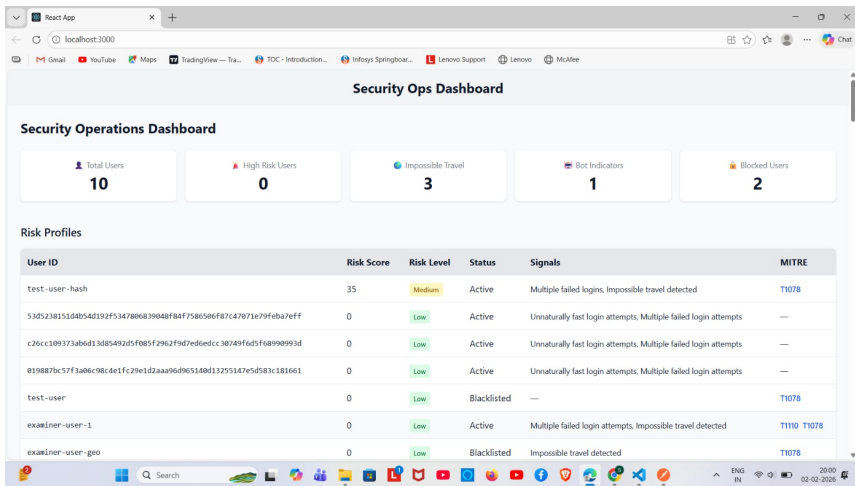
- **Risk Score Visualization**

- Risk score charts display user risk trends over time.

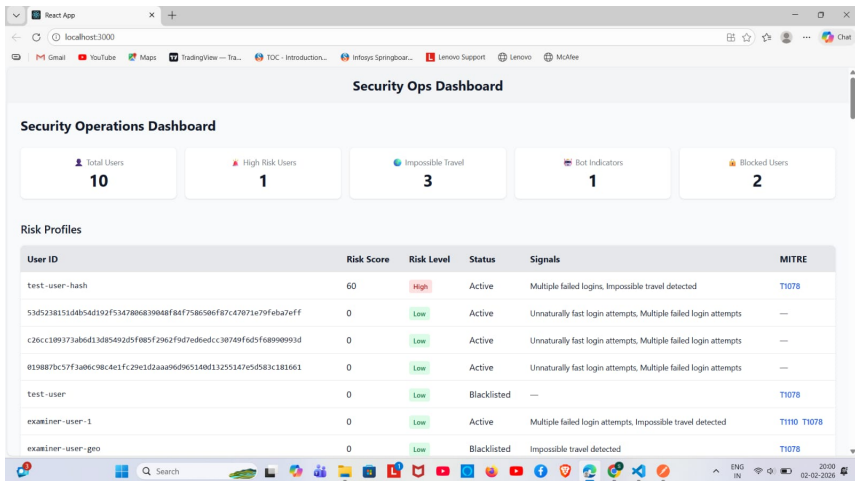
- **Alerts and Indicators**

- Alerts highlight suspicious logins, bot activity, and impossible travel events.

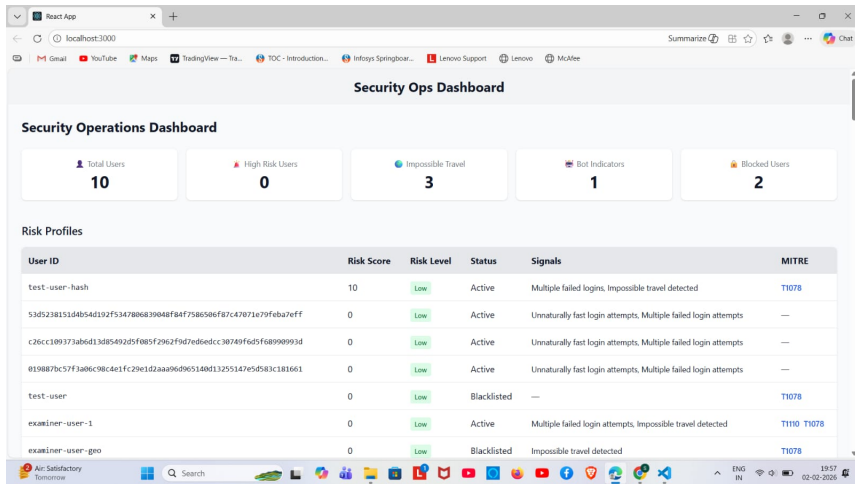




Output (contd.)



Output (contd.)



Expected outcomes

- A secure admin authentication system implementing username–password verification, JWT-based access control, and multi-factor authentication.
- Controlled access through IP whitelisting, reducing unauthorized and untrusted login attempts.
- Centralized logging of all login activities to support monitoring, auditing, and security analysis.
- Detection of suspicious behavior such as failed logins, MFA failures, bot activity, and unusual location changes.



Expected outcomes (contd.)

- A dynamic risk scoring mechanism that evaluates login behavior and identifies high-risk users or IP addresses.
- A real-time security dashboard displaying alerts, risk scores, and security metrics for proactive monitoring and response.



Conclusion

- The project successfully implements a secure admin authentication workflow using JWT, multi-factor authentication, IP whitelisting, and risk-based access control.
- Real-time monitoring and analysis of login activities enable effective detection of suspicious behavior and automated security actions.
- The interactive security dashboard provides clear visibility into system security, making the solution suitable for academic and small-scale security implementations.



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